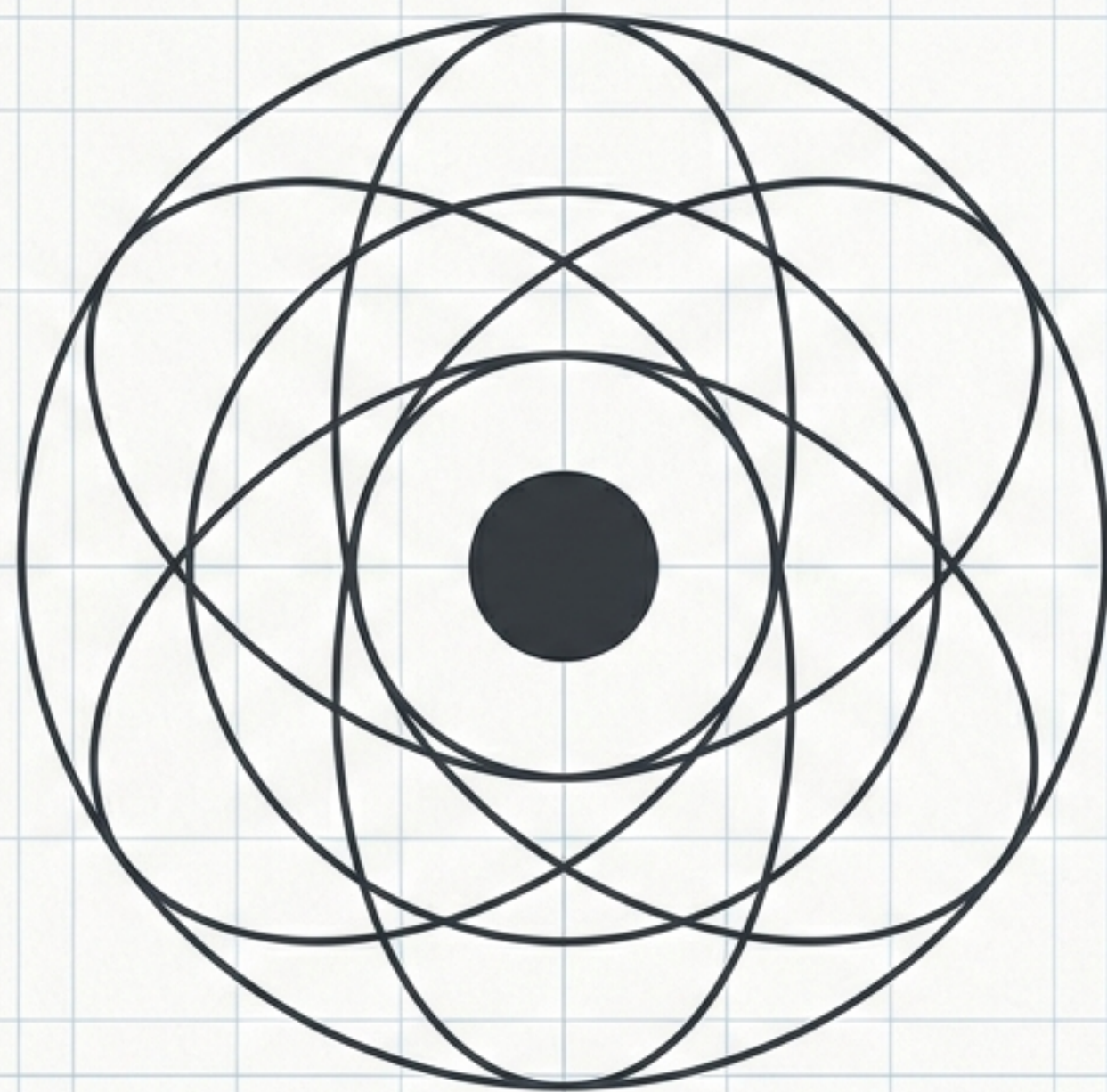




The Language of Chemistry

A Typographic Blueprint to Symbols, Valency, and Equations

The Foundational Entities of Matter

The Particle Identity Matrix

	Atom	Molecule	Ion	Free Radical ●
Independent Existence	Rarely (except noble gases)	Yes	In solution or lattice	Yes (short-lived)
Electrical Charge	Neutral	Neutral	Positive (Cation) Negative (Anion)	Neutral
Stability & Reactivity	Seeks octet configuration	Stable (completed octets)	Stable electron configuration	Extremely unstable and aggressive

The Alphabet: Designing Chemical Symbols

The Evolution of Notation

Alchemists
(Pre-1600)



Earth



Silver

Dalton
(1808)



Hydrogen

Berzelius
(Modern IUPAC)

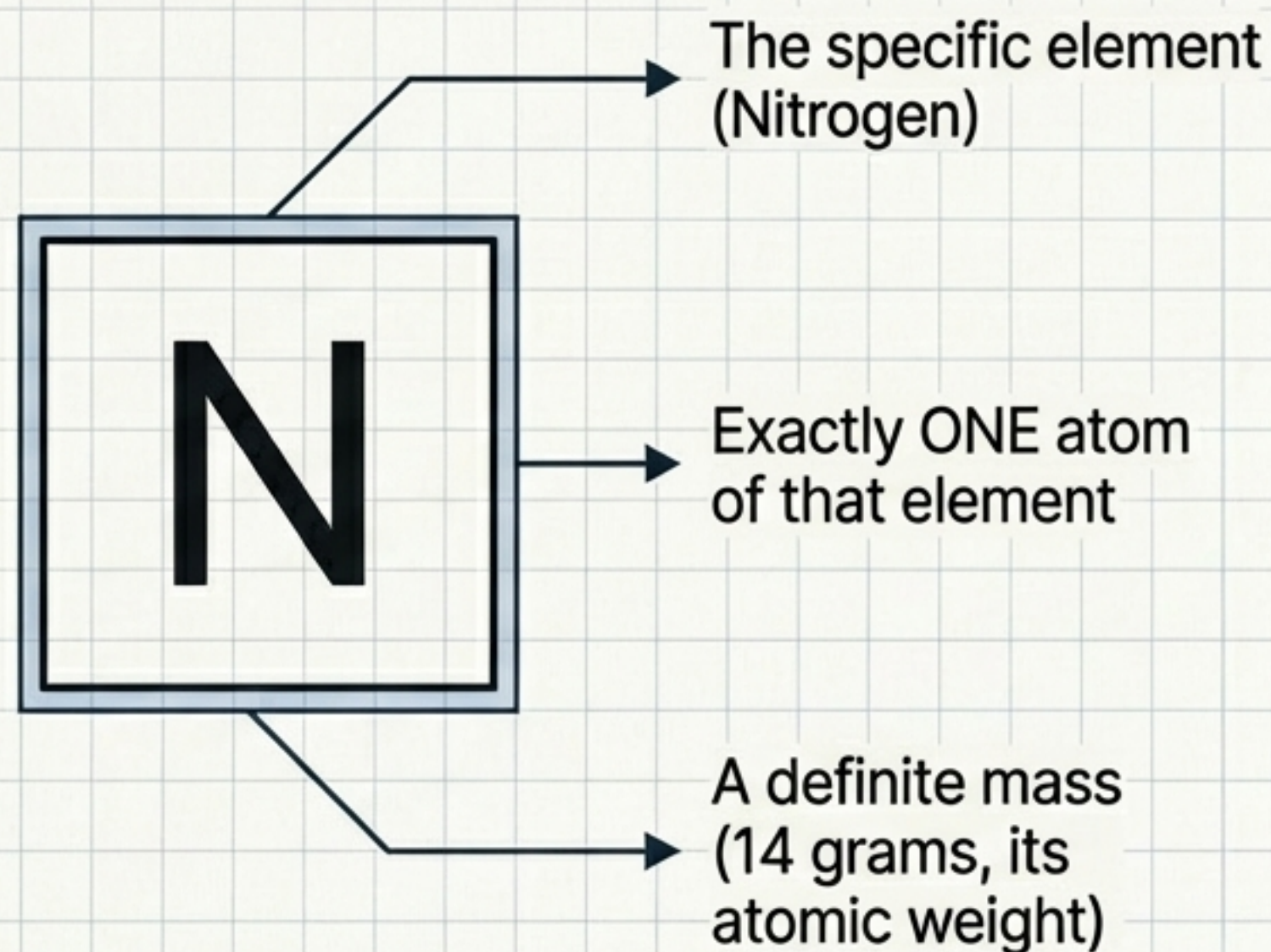


Carbon

Latin Derivations

Gold: Au (Aurum)	Iron: Fe (Ferrum)
Lead: Pb (Plumbum)	Tungsten: W (Wolfram)

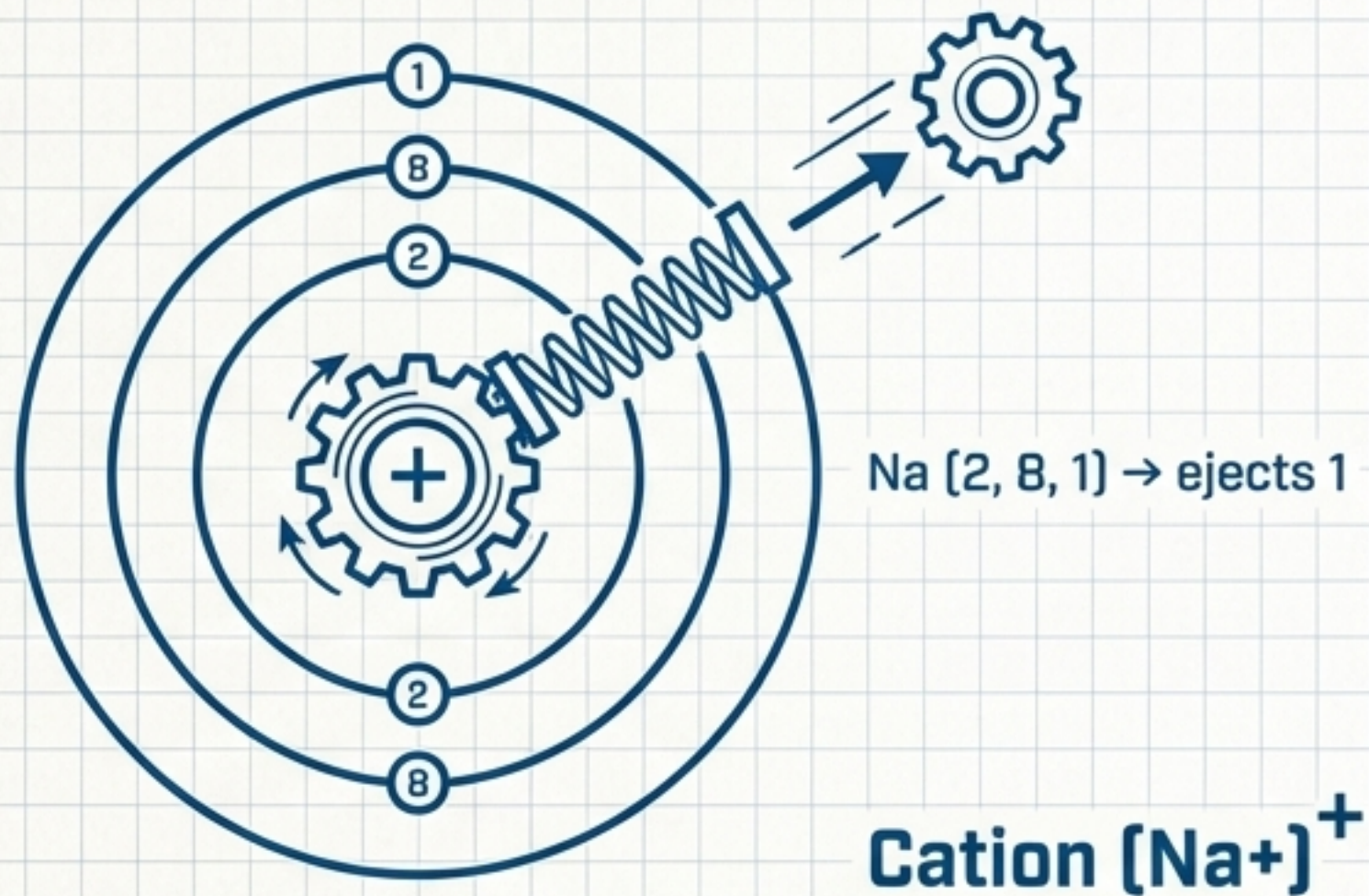
Significance of a Symbol



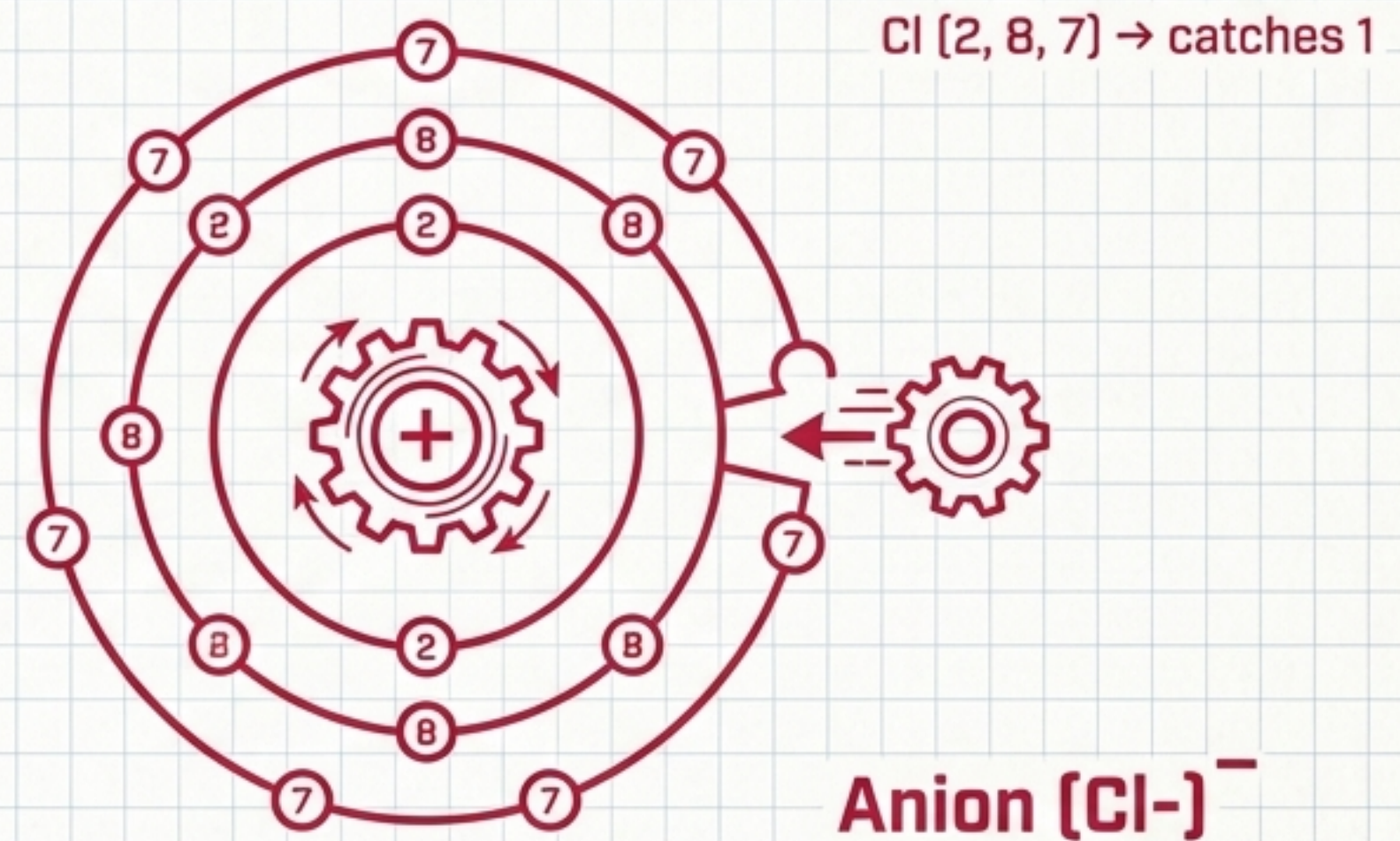
Valency: The Mechanics of Combination

The number of electrons an atom can lose, gain, or share to attain an 8-electron (octet) stability.

Metals (Electron Donors)

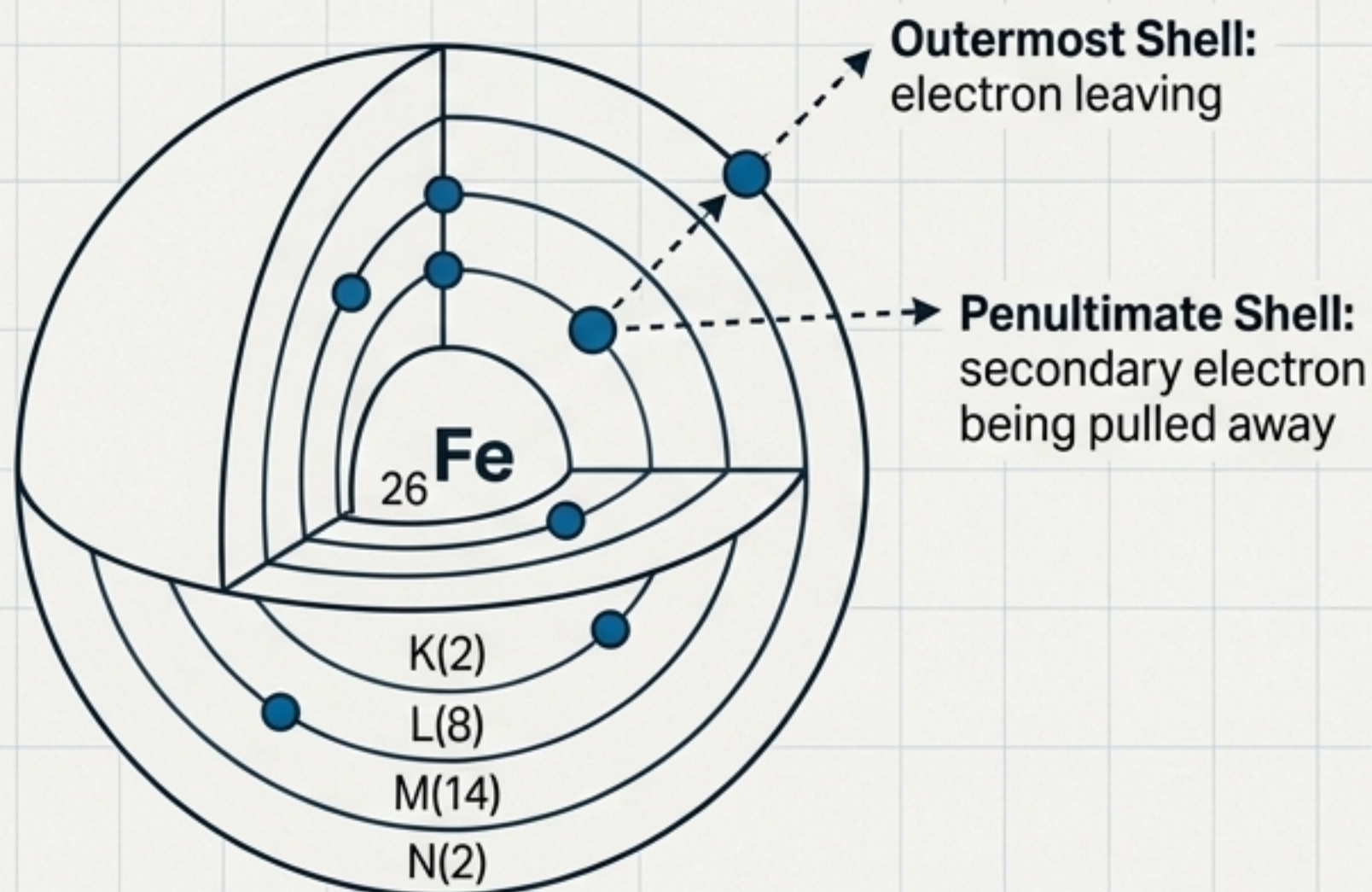


Non-metals (Electron Acceptors)



Variable Valency and the Penultimate Shell

When an element loses electrons not just from the outermost (valence) shell, but also from the penultimate (next to outermost) shell.



Stock Notation Rules

Lower valency = -ous suffix | Higher valency = -ic suffix

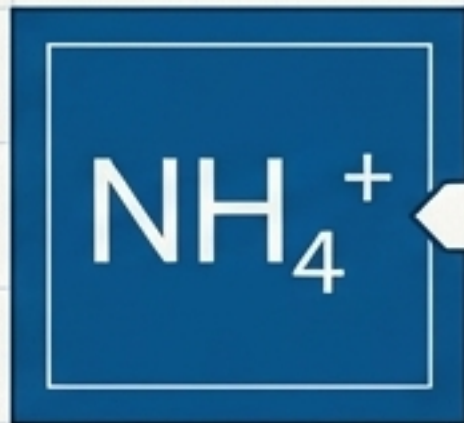
Element & Valency		Compound Examples & Notation
Iron (Fe)	Fe(II) or Ferrous (Fe^{2+})	FeO , FeCl_2 , Fe(OH)_2 , FeSO_4
	Fe(III) or Ferric (Fe^{3+})	Fe_2O_3 , FeCl_3 , $\text{Fe}_2(\text{SO}_4)_3$, Fe(OH)_3
Copper (Cu)	Cu(I) or Cuprous (Cu^{1+})	Cu_2O , CuCl
	Cu(II) or Cupric (Cu^{2+})	CuO , CuCl_2 , Cu(OH)_2 , CuSO_4

Radicals: The Modular Blocks of Compounds

A simple or complex structure behaving as a single unit with a net charge.

Basic Radicals (Electropositive Cations)

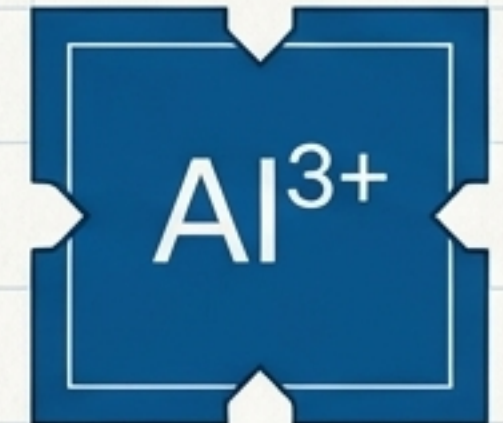
Monovalent:



Divalent:

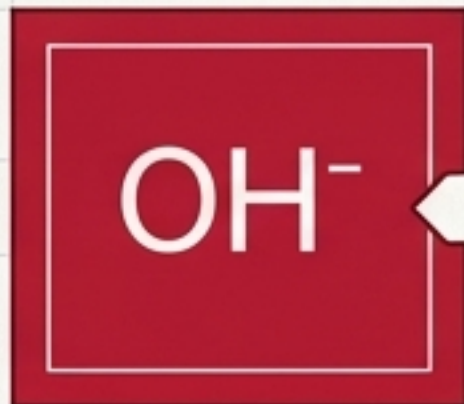


Trivalent:

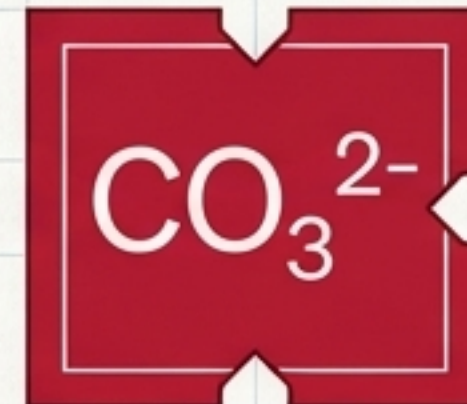
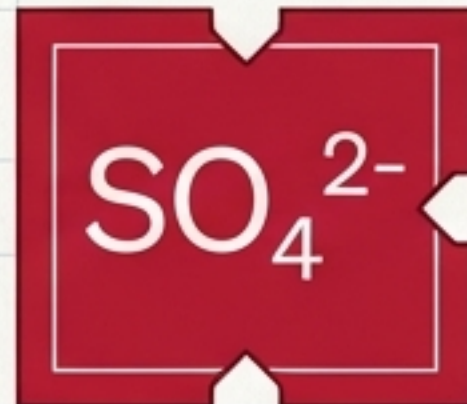


Acidic Radicals (Electronegative Anions)

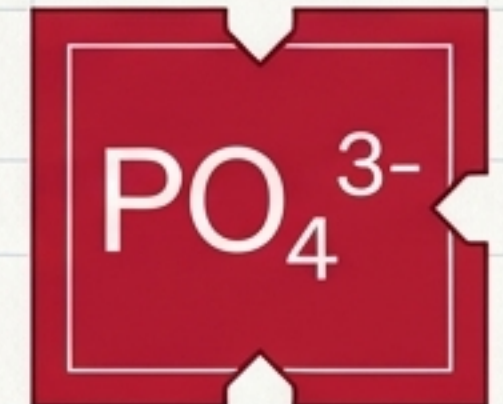
Monovalent:



Divalent:

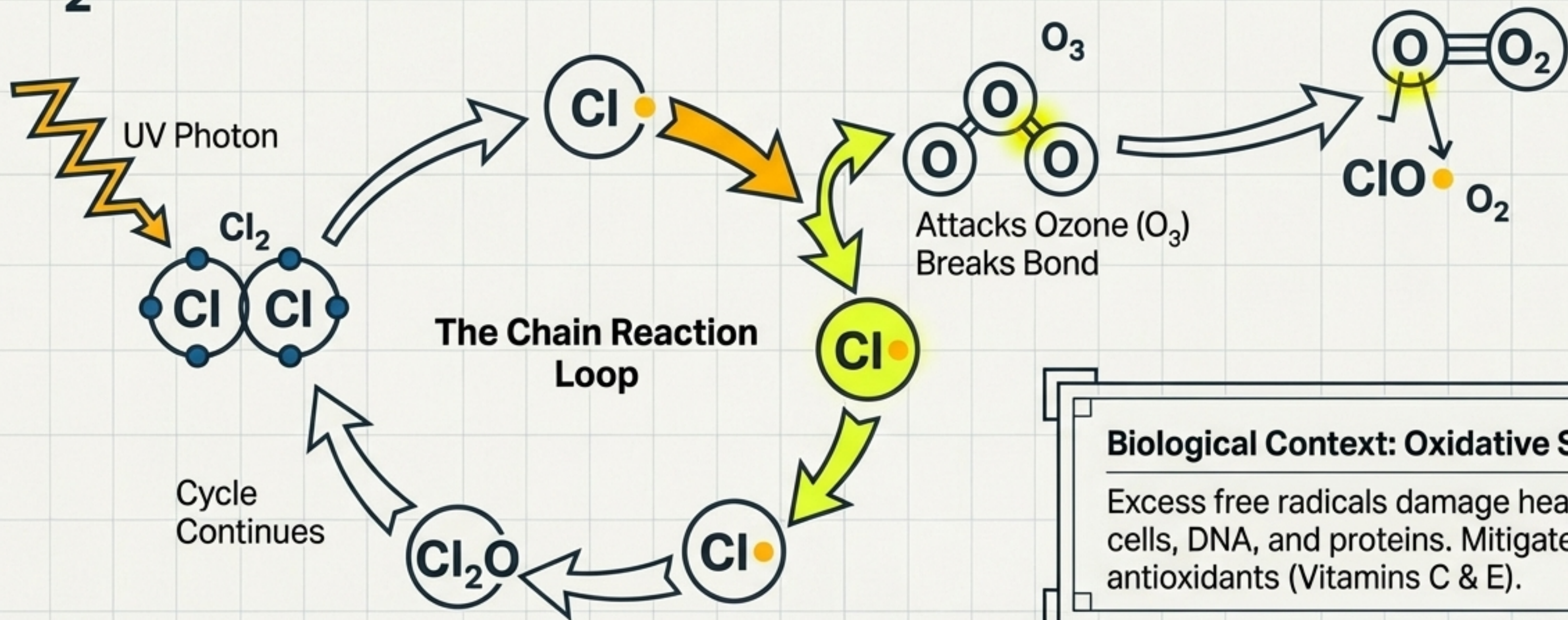


Trivalent:



Free Radicals: The Rogue Particles

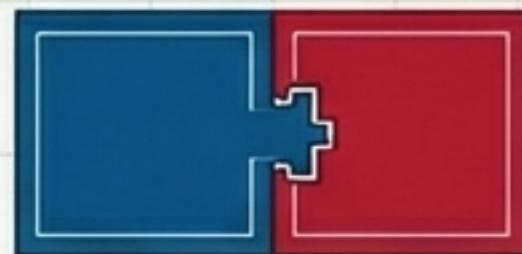
Atoms or molecules containing an unpaired valence electron, formed via homolytic fission. Exceptionally reactive.



Biological Context: Oxidative Stress

Excess free radicals damage healthy cells, DNA, and proteins. Mitigated by antioxidants (Vitamins C & E).

Writing Formulas: The Criss-Cross Method



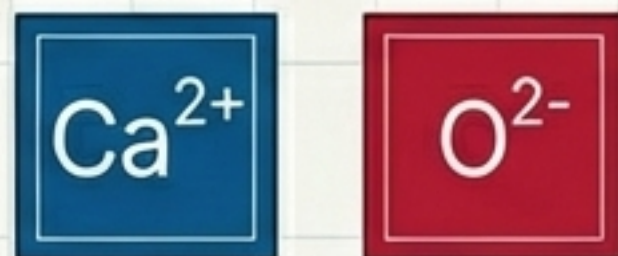
Step 1: Alignment

Place symbols side-by-side.



Step 2: Valency Tags

Write valencies on top.



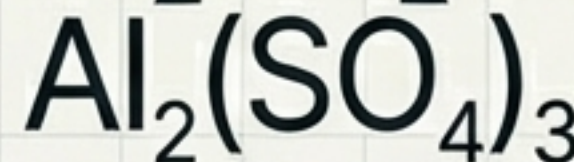
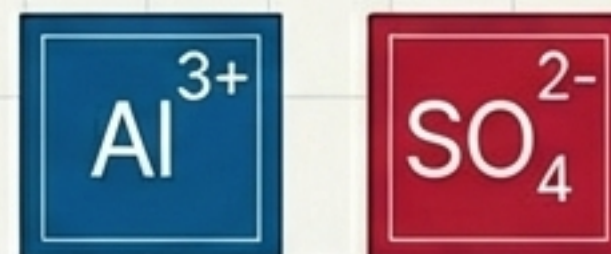
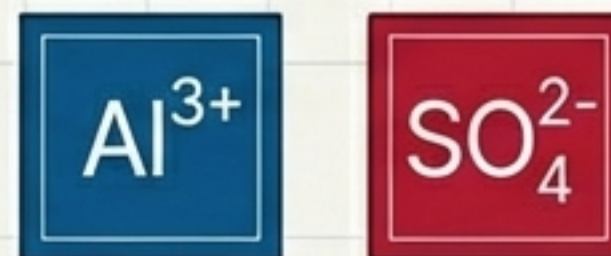
Step 3: Simplify

Divide by Highest Common Factor.



Step 4: Criss-Cross

Valencies slide diagonally down to become subscripts. Parenthesis enclose complex radicals > 1.



Step 1: Alignment

Place symbols side-by-side.

Step 2: Valency Tags

Write valencies on top.

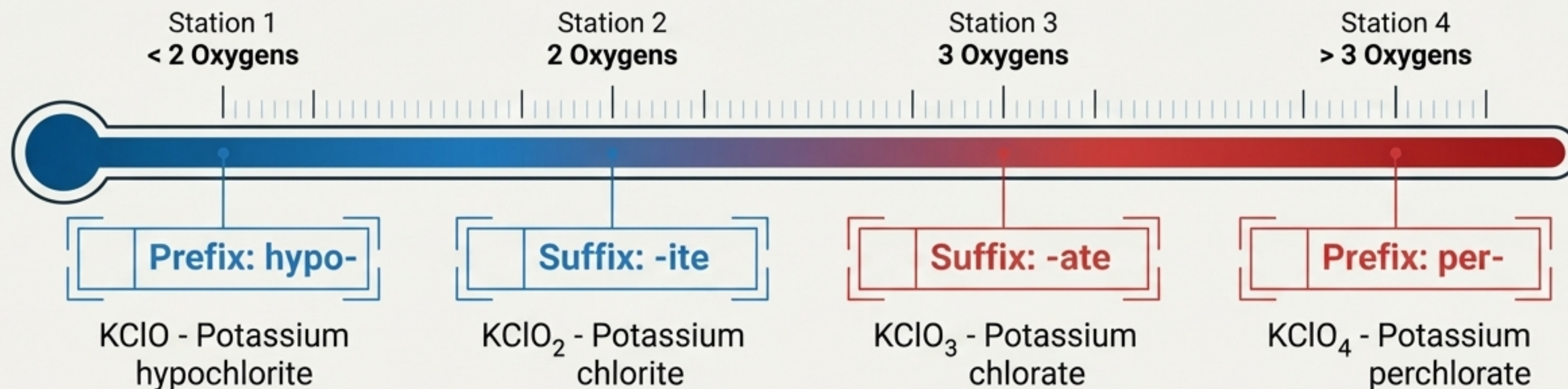
Step 3: Simplify

Divide by Highest Common Factor. (No change)

Step 4: Criss-Cross

Valencies slide diagonally down to become subscripts. Parenthesis enclose complex radicals > 1.

Nomenclature: The Oxygen Spectrum



Naming Acids

- Binary acids: **hydro-** + **-ic** (HCl)
- Fewer Oxygen: **-ous** (H₂SO₃ Sulphurous acid)
- More Oxygen: **-ic** (H₂SO₄ Sulphuric acid)

The Evolution of a Chemical Sentence

The Equation Evolution Table

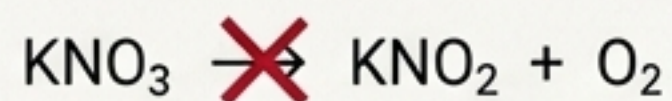
Level 1: Word Equation

Qualitative change, but lacks molecular truth
Carbon + Oxygen $\rightarrow$ Carbon dioxide



Level 2: Skeletal Equation

Correct symbols, but violates conservation laws

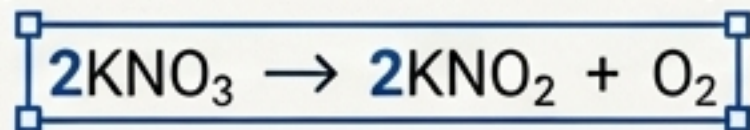


⚠ Atoms are unequal



Level 3: Balanced Equation

Quantitatively accurate. Number of atoms is perfectly equal.



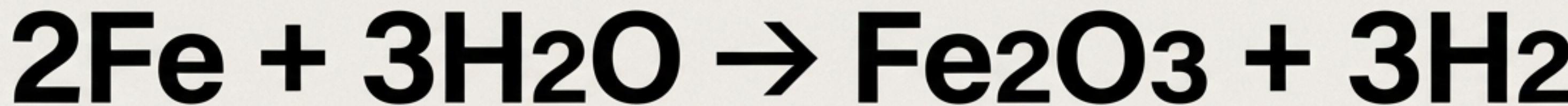
Why Balance?

The **Law of Conservation of Matter**: Matter cannot be created or destroyed, only rearranged.

Balancing Mechanics: Hit and Trial

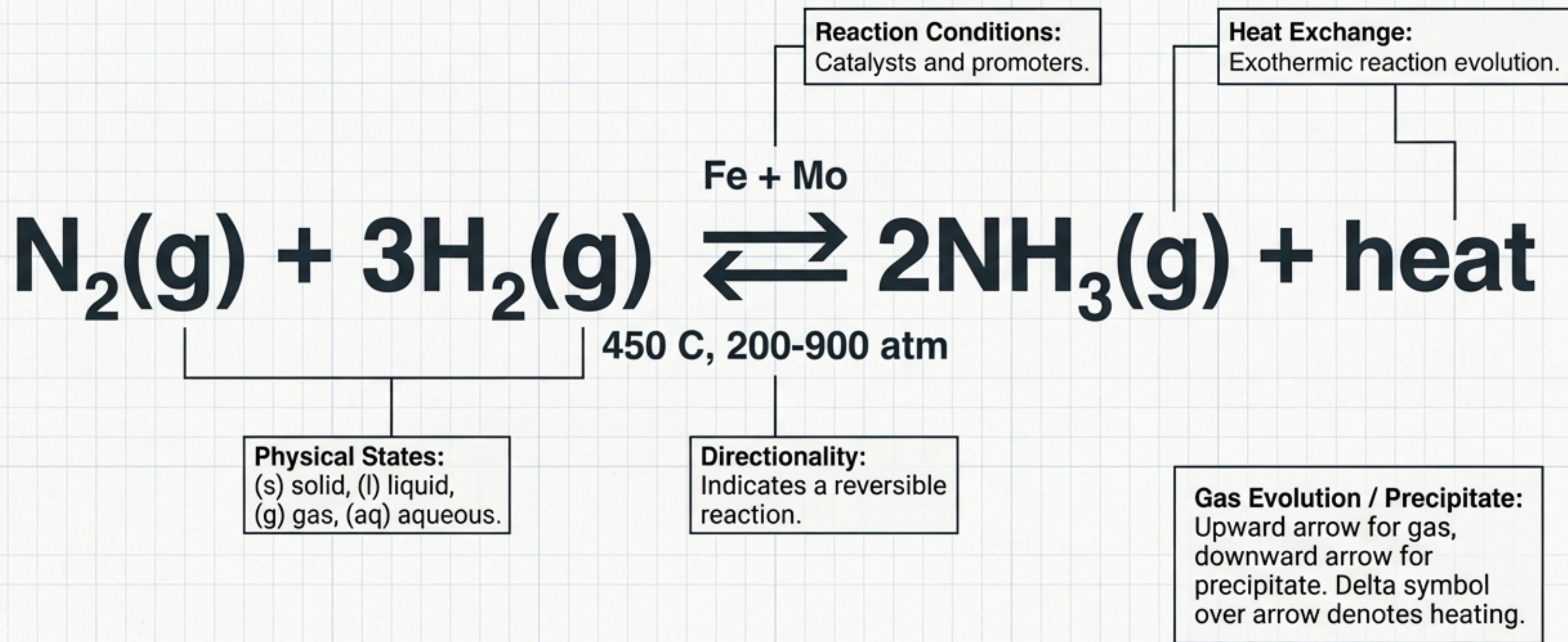


Step 1: Initial Tally		Step 2: Balance Metals		Step 3: Balance Oxygen		Step 4: Balance Hydrogen	
Fe		2Fe		3H₂O		3H₂	
Reactant Side	Product Side	Reactant Side	Product Side	Reactant Side	Product Side	Reactant Side	Product Side
Fe = 1 H = 2 O = 1	Fe = 2 H = 2 O = 3	Fe = 2 H = 2 O = 1	Fe = 2 H = 2 O = 3	Fe = 2 H = 6 O = 3	Fe = 2 H = 2 O = 3	Fe = 2 H = 6 O = 3	Fe = 2 H = 6 O = 3
✗		→		→		✓	
Atoms are unequal							

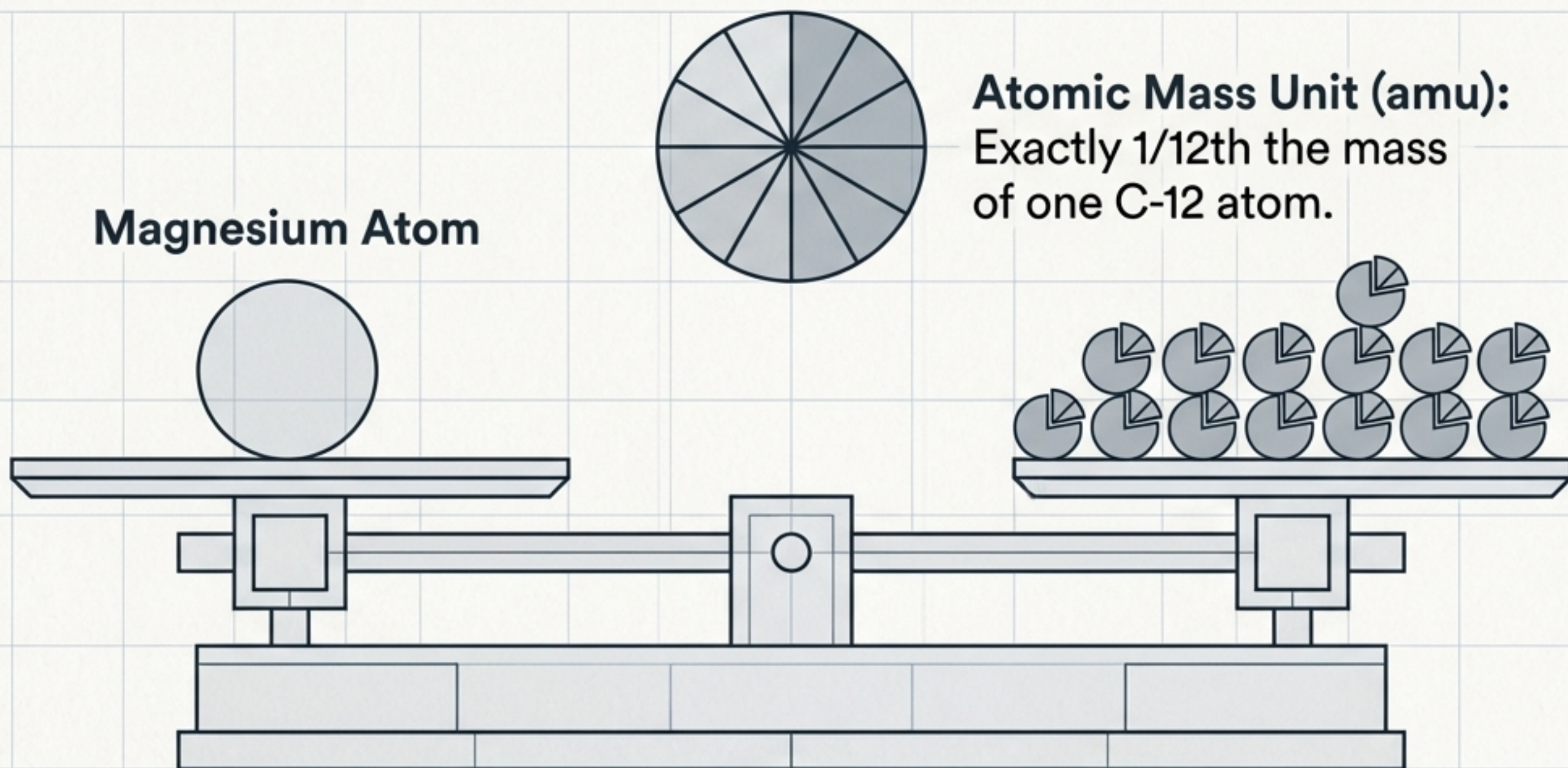


↑
Coefficients multiply the whole molecule;
subscripts are locked structural elements.

Contextual Syntax: The Informative Equation



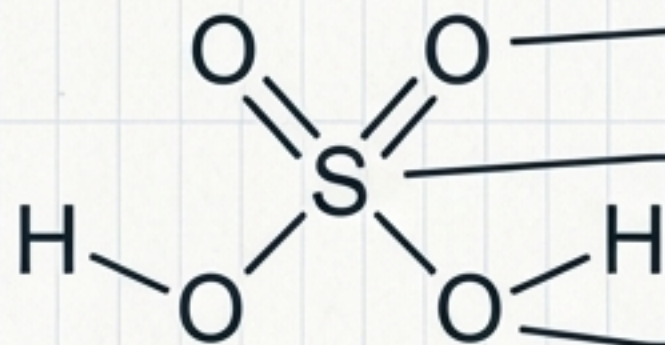
Weighing the Invisible: Relative Atomic Mass



The mass of atoms is too infinitesimally small to use standard weights. R.A.M. is a relative ratio representing how many times heavier one atom is compared to the $\frac{1}{12}$ th C-12 standard.

The Geometry of Mass: Relative Molecular Mass

Calculation: **RMM of Sulphuric Acid (H_2SO_4)**



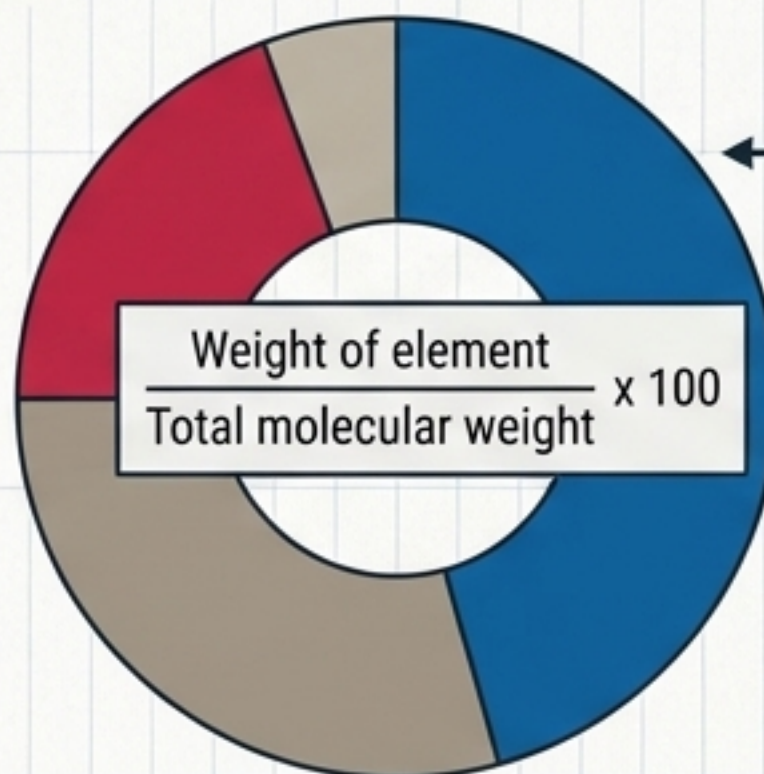
2 atoms (Mass: 1) $\rightarrow 2 \times 1 = 2$

1 atom (Mass: 32) $\rightarrow 1 \times 32 = 32$

4 atoms (Mass: 16) $\rightarrow 4 \times 16 = 64$

$2 + 32 + 64 = 98 \text{ amu}$

Percentage Composition (Urea: NH_2CONH_2)

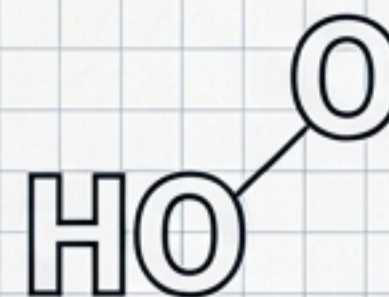
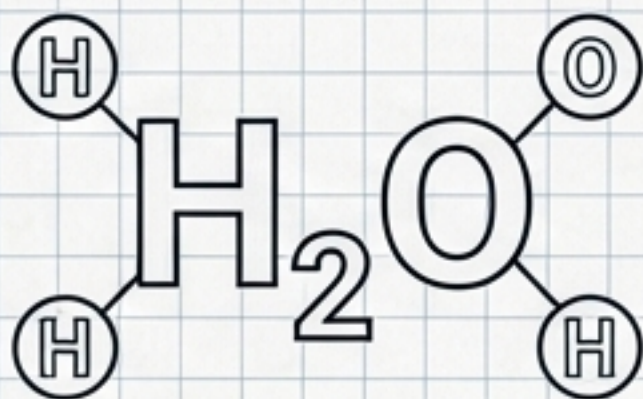


Total Mass = 60 amu.
Nitrogen mass = 28 amu.
Nitrogen Percentage = 46.6%

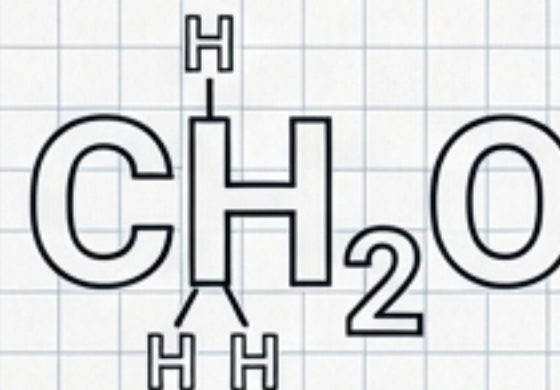
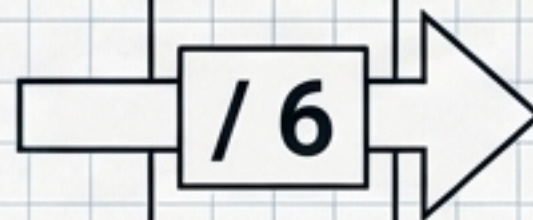
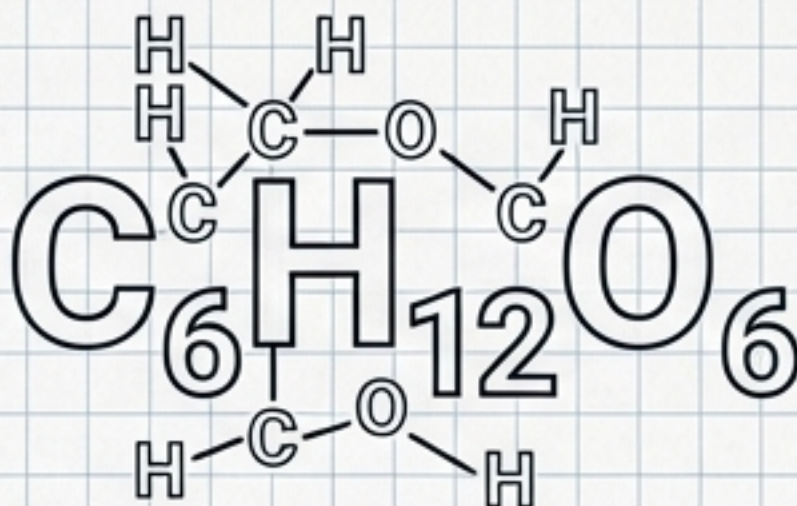
Empirical vs. Molecular Formulas

Molecular Formula (The Full Blueprint)

The actual number of atoms of each element present.



Ratio 1:1



Ratio 1:2:1

Key Takeaway: Empirical formulas can be identical for entirely different compounds (e.g., Glucose and Acetic acid both share the CH_2O ratio).

Synthesis: Reading the Complete Language

Layer 1: Qualitative (The Words)

Calcium carbonate reacts with hydrochloric acid to produce calcium chloride, water, and carbon dioxide.

Layer 2: Particulate (The Grammar)

One molecule of CaCO_3 reacts with two molecules of HCl to yield...



100g

73g

111g

18g

44g /
22.4 Litres
at STP

Layer 3: Macroscopic (The Comprehension)

The language of chemistry seamlessly scales from the invisible interactions of a few atoms directly to real-world, measurable quantities.